

LOGANALYZER – User Commands Manual

NAME

loganalyzer - analyze a log file and report basic statistics

SYNOPSIS

loganalyzer -f <logfile>

DESCRIPTION

loganalyzer reads a log file and counts the total number of lines.

It also counts how many lines contain the keywords ERROR, WARNING, and INFO.

The program uses mmap() for efficient file access. This allows the file to be mapped directly into memory instead of reading it line by line using normal read calls.

OPTIONS

-f <logfile> Specifies the log file to analyze.

EXAMPLES

```
./loganalyzer -f sample.log
```

Log Analysis Result

File name : sample.log

Total lines : 8

ERROR count : 2

WARNING count : 2

INFO count : 4

SYSTEM CALLS USED

open() - opens the log file

fstat() - gets file information and file size

mmap() - maps the file into memory

munmap() - releases the mapped memory

close() - closes the file descriptor

getopt() - parses command-line arguments

ERROR HANDLING

Handles missing arguments

Handles file not found

Handles empty file

Handles mmap failure